Nitin Kukreja

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Summary

AI engineer (5+ yrs) who builds **and** ships LLMs in a regulated, national-scale setting: post-trained and RL-tuned NPCI's 24B financial LLM (FiMI) and deployed it as the agentic assistant behind UPI Help / MyUPI. Named author on 3 arXiv papers, including the IndicBankBench safety benchmark.

Experience

National Payments Corporation of India (NPCI) Mumbai, India
Senior Associate Data Scientist *Apr 2025 – Present*

FiMI – 24B domain-specific financial LLM (arXiv:2602.05794)

- Led post-training and agentic integration of FiMI, now powering UPI Help in production; achieved **+20%** over Mistral Small 24B base on finance reasoning and **+87%** on domain tool calling.
- Designed multi-stage training (CPT → IFT → SFT) on **68B** curated English/Hindi/Hinglish tokens, distilled from a 30T-token corpus via PII anonymisation, deduplication and semantic validation; generated synthetic tool-calling and multi-turn SFT data.
- Ran distributed training on **64 H100** GPUs with DeepSpeed ZeRO-3 and FlashAttention-2 (Slurm, Docker, Kubernetes).

FiMI Banking – RL post-training & evaluation for safe bank agents (arXiv:2609.03960)

- Applied preference optimisation and verifiable-reward RL to multi-turn banking tool use: out-of-scope refusal **52%→80%**, edge-case score 0.509→0.718, order-sensitive 0.590→0.679, with **29% fewer tokens**.
- Built **Banking Gym**, a deterministic NeMo Gym-compatible RL environment: stateful bank simulator, confirmation-gated writes, verifier-graded rewards and parallel isolated rollouts.
- Co-authored **IndicBankBench** (arXiv:2609.29167): 799 multi-turn banking cases across 20 axes with deterministic safety and tool-use gates; evaluated 11 models from 8 families.

UPI Help & MyUPI – FiMI-powered agentic assistant (production, ∼100K DAU; launched at GFF 2025)

- Architected a swarm-based multi-agent backend (LangGraph, MCP, FastAPI) for dispute resolution (UDIR), mandate management and escalation in 5 languages, with guardrails and hallucination detection (faithfulness checks, LLM-as-judge).
- Built RAG and Graph-RAG over UPI policy (Qdrant, Neo4j); served models with vLLM and Triton.

Associate Data Scientist *Sep 2023 – Mar 2025*

- Built GNN fraud models (PyTorch) scoring **15,000 transactions/sec at 20 ms latency**, contributing to **$1M+** annual fraud-loss reduction; graph-traversal AML workflows cut manual review effort by **80%**.
- Contributed to real-time anomaly detection on NVIDIA DGX, featured by NVIDIA at SuperComputing '24.

Xoriant Solutions Pune, India
Software Engineer (Associate Software Engineer until Aug 2022) *Aug 2021 – Sep 2023*

- Built an enterprise GPT-3.5 chatbot (LangChain, Azure Cognitive Search), cutting a client's support workload by **40%**.
- Shipped document AI: LiLT/BERT entity extraction at **95%** accuracy, doc2vec/Keras classification, spaCy + BERT NER with Azure Form Recognizer.

Publications

IndicBankBench: Evaluating Safety and Reliability of Language Model Assistants in Indian Retail Banking – arXiv:2609.29167, 2026; code: github.com/npci/IndicBankBench

FiMI Banking: A Sovereign Model for Indian Retail Banking – arXiv:2609.03960, 2026

FiMI: A Domain-Specific Language Model for Indian Finance Ecosystem – arXiv:2602.05794, 2026

Technical Skills

LLM & Agents: SFT/CPT/IFT, PEFT/LoRA, RL (RLVR, GRPO, preference optimisation), evaluation & benchmarks, RAG/Graph-RAG, LangGraph, LangChain, MCP, guardrails
ML & Infra: Python, PyTorch, Hugging Face, GNNs, DeepSpeed, FlashAttention-2, Slurm, vLLM, Triton, Docker, Kubernetes, MLflow
Data & Cloud: SQL, MongoDB, Neo4j, Qdrant, FAISS, FastAPI, Azure OpenAI, AWS (SageMaker, Bedrock), GCP

Education

R. C. Patel Institute of Technology Shirpur, India
B.Tech, Computer Science & Engineering – CGPA 8.4/10 *2017 – 2021*